Topic 4: File Sharing in Windows Server

Task 1: Question: On your Windows Server VM, create a new folder named 'MusicShare' on the D: drive and share it with the network using the Share Wizard.

The server had no D: drive , so a new virtual disk was added first. The folder was then created on it and shared.

1. Shut down Server01, open VirtualBox Settings > Storage and add a new virtual hard disk (VDI, dynamically allocated).
2. Start Server01, open Disk Management, initialize the new disk (GPT) and create a New Simple Volume formatted as NTFS.
3. Change the CD drive letter so that the new volume could take D:, then assign D: to the new volume.
4. Create the folder D:\MusicShare.
5. Right-click MusicShare > Properties > Sharing and share the folder with the network (File Sharing wizard, then Advanced Sharing to fine-tune the share permissions).

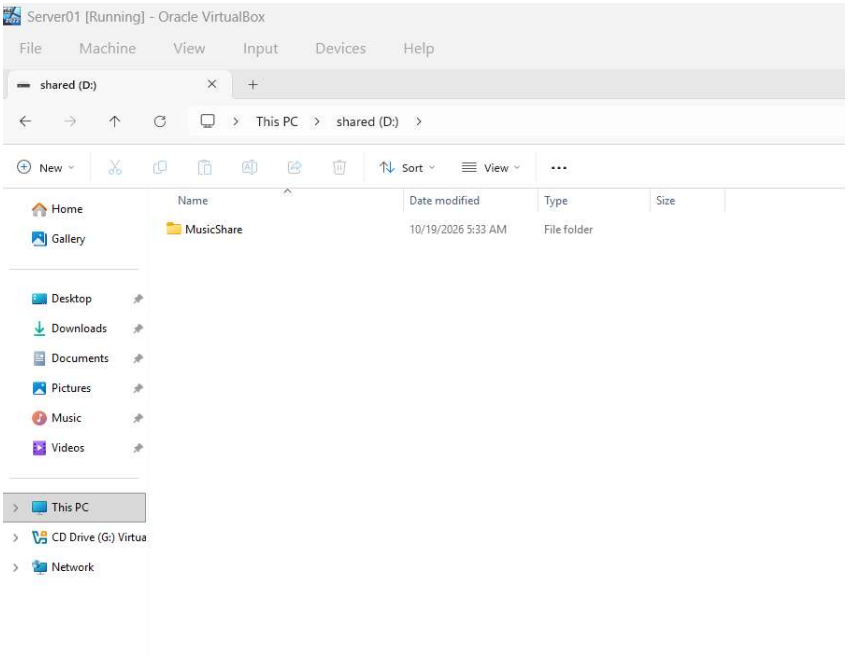


Figure 1: MusicShare folder on the new D: drive ("shared") of Server01

Task 2: Question: Set NTFS permissions on the 'MusicShare' folder so that the user 'Student1' has Read & Write access, while 'Student2' has Read-only access.

NTFS permissions were set on the Security tab of D:\MusicShare. Inherited entries from the drive root were removed so that no extra group could give Student 2 write access.

User	NTFS permission set	Effect
Student 1	Modify (includes Read & execute, List folder contents, Read, Write)	Read & Write

User	NTFS permission set	Effect
Student 2	Read & execute, List folder contents, Read	Read-only
Administrators	Full control (default)	Administration

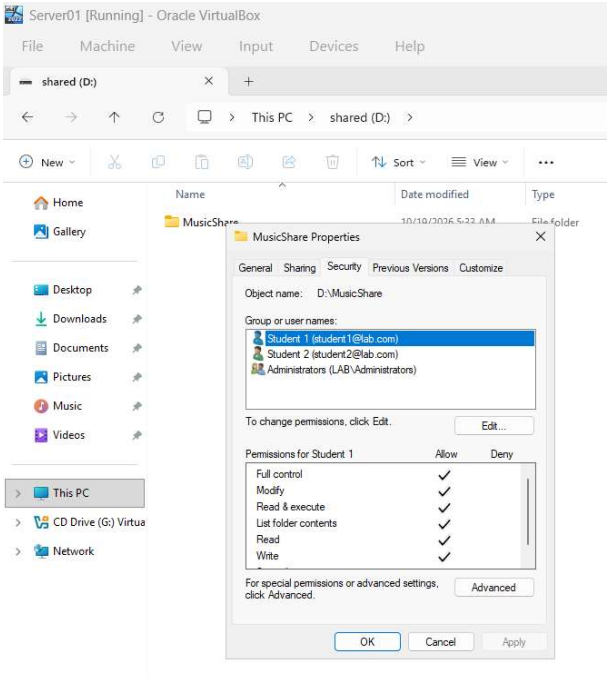


Figure 2: Security tab, Student 1 has Modify and Write (Allow)

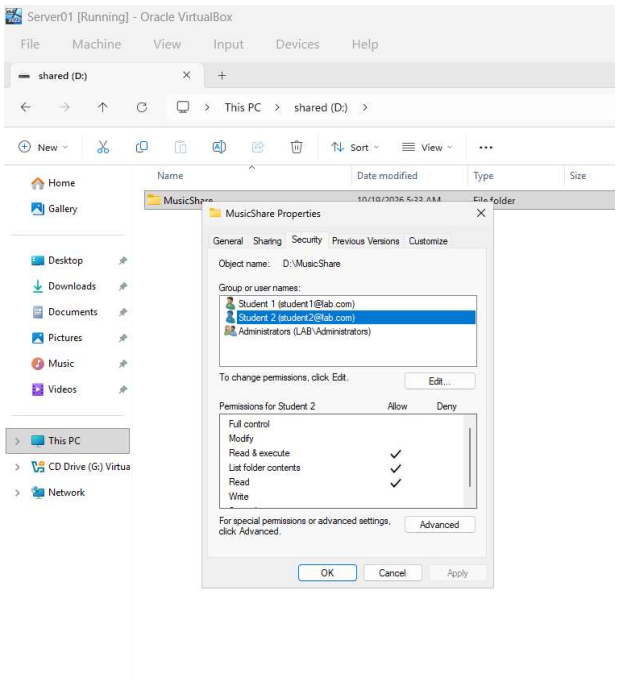


Figure 3: Security tab, Student 2 has only Read & execute, List folder contents and Read

In plain language: *Student 1 is allowed to read, add and change files. Student 2 is only allowed to look at and open files. Nobody else has been left on the list, so there is no hidden way for Student 2 to gain more rights.*

Task 3: Configure Share permissions for 'MusicShare' so that only members of the 'MusicLovers' group can access the share, and all others are denied.

A security group named MusicLovers was created in Active Directory Users and Computers and both students were added as members. In Advanced Sharing > Permissions, the individual users and Everyone were removed and only LAB\MusicLovers was added, with Allow for Change and Read.

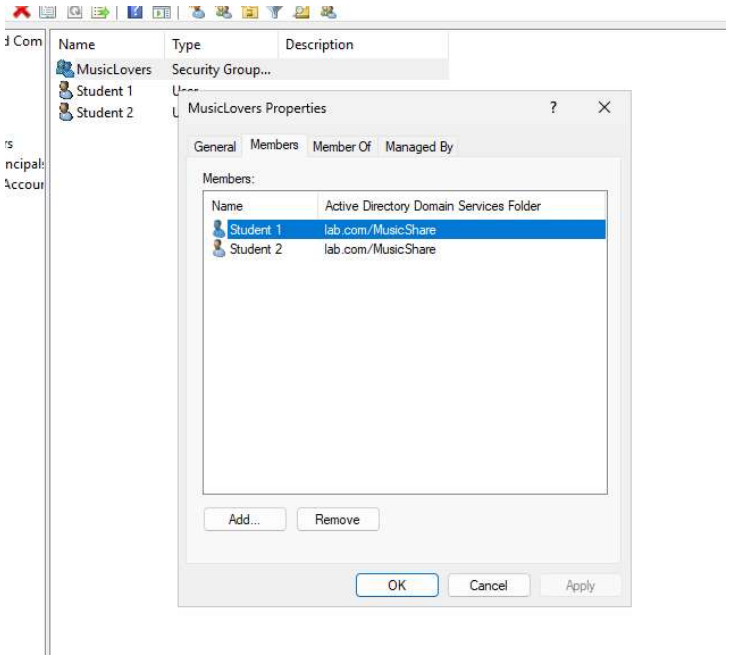


Figure 4: MusicLovers group Members tab (Student 1 and Student 2)

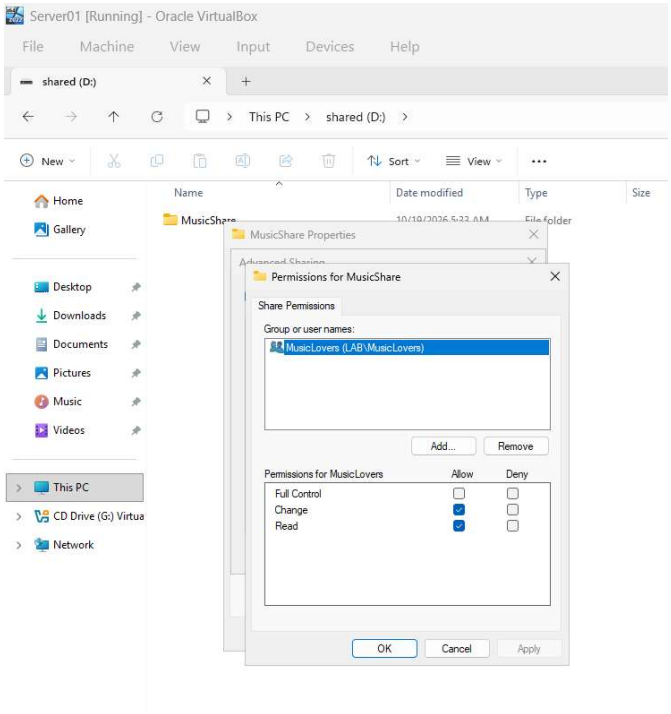


Figure 5: Share permissions for MusicShare, only LAB\MusicLovers (Change, Read)

Anyone who is not on the share permission list is denied automatically, so no explicit Deny entry is needed. Change (not Full Control) is enough at the share level because the NTFS permissions do the detailed restricting.

Task 4: Question: Simulate these access scenarios: log in as 'Student1' and try to copy a file into 'MusicShare', then log in as 'Student2' and attempt to delete a file from the same folder. Write down what happens in each case and explain why based on the permissions set.

Answer: Both tests were run from the client PC (client01) over the network path \ \SERVER01\MusicShare, which was mapped as a network drive. Before each test, whoami was run in Command Prompt to prove which user was logged in.

```
C:\Users\student1> whoami
lab\student1
```

Scenario A: Student 1 copies a file into MusicShare

What happened: the copy succeeded. The file "copyfile" appeared in the share and is also still on the Desktop, which shows it was copied, not moved.

Why: the share permission (MusicLovers: Change) lets Student 1 through the front door, and the NTFS permission (Student 1: Modify) allows writing. Both layers allow the action.

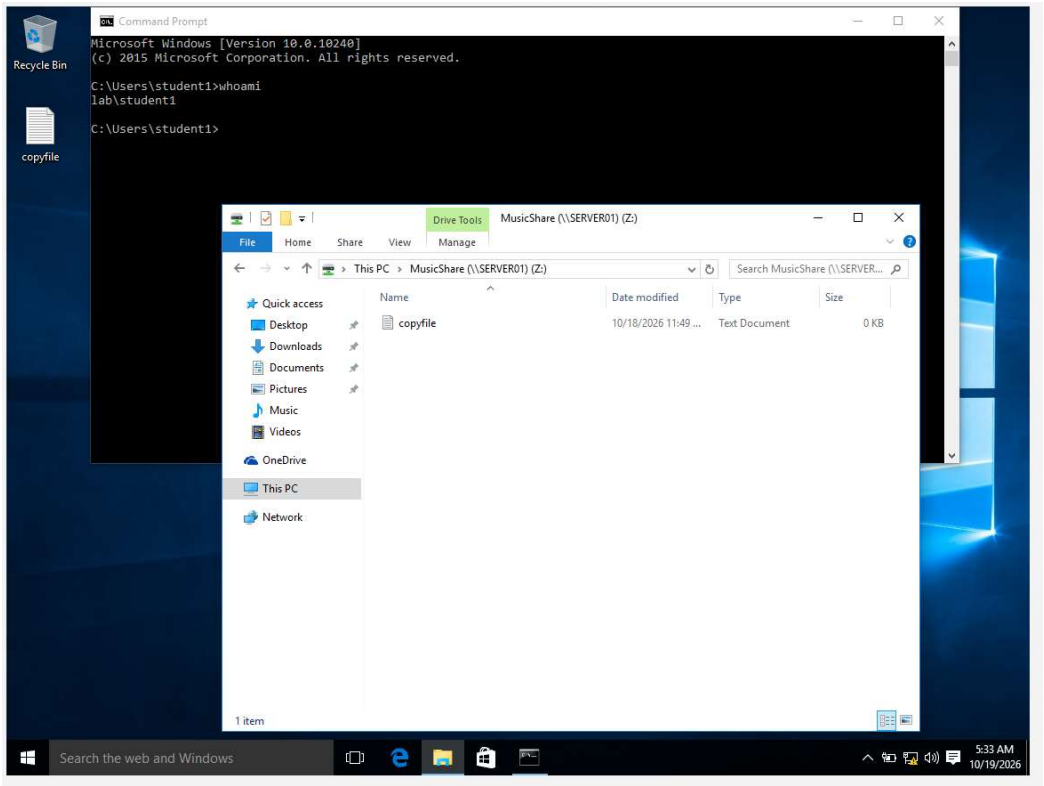


Figure 6: Logged in as lab\student1 (whoami); copyfile copied into \\SERVER01\MusicShare

Scenario B: Student 2 attempts to delete a file from MusicShare

```
C:\Users\student2> whoami
lab\student2
```

What happened: Windows showed "File Access Denied: You need permission to perform this action". The file was not deleted.

Why: Student 2 passed the share layer because they are a member of MusicLovers, which has Change. However, the NTFS permission for Student 2 is Read-only, with no Write or Modify, and deleting needs Modify. Windows applies the more restrictive permission, so the delete was refused.

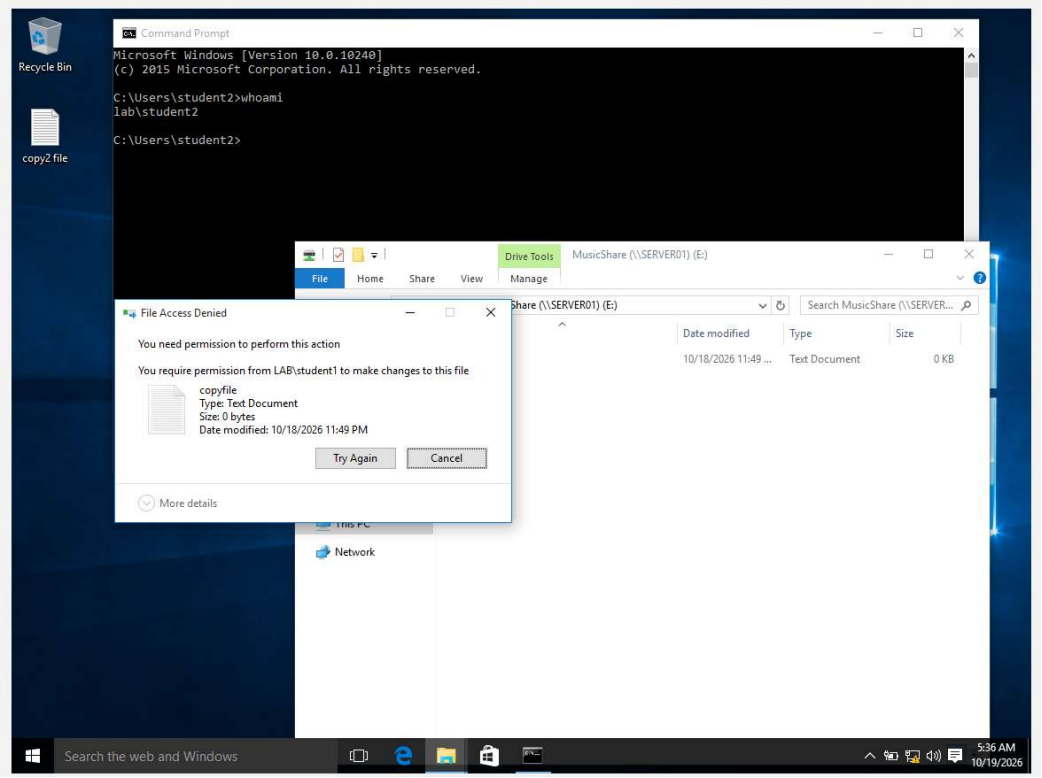


Figure 7: Logged in as lab\student2 (whoami); delete of copyfile denied

Extra test: Student 2 tries to copy a file into MusicShare

What happened: "Destination Folder Access Denied: You need permission to perform this action". Student 2 cannot add files either, for the same reason: NTFS gives Student 2 no Write permission.

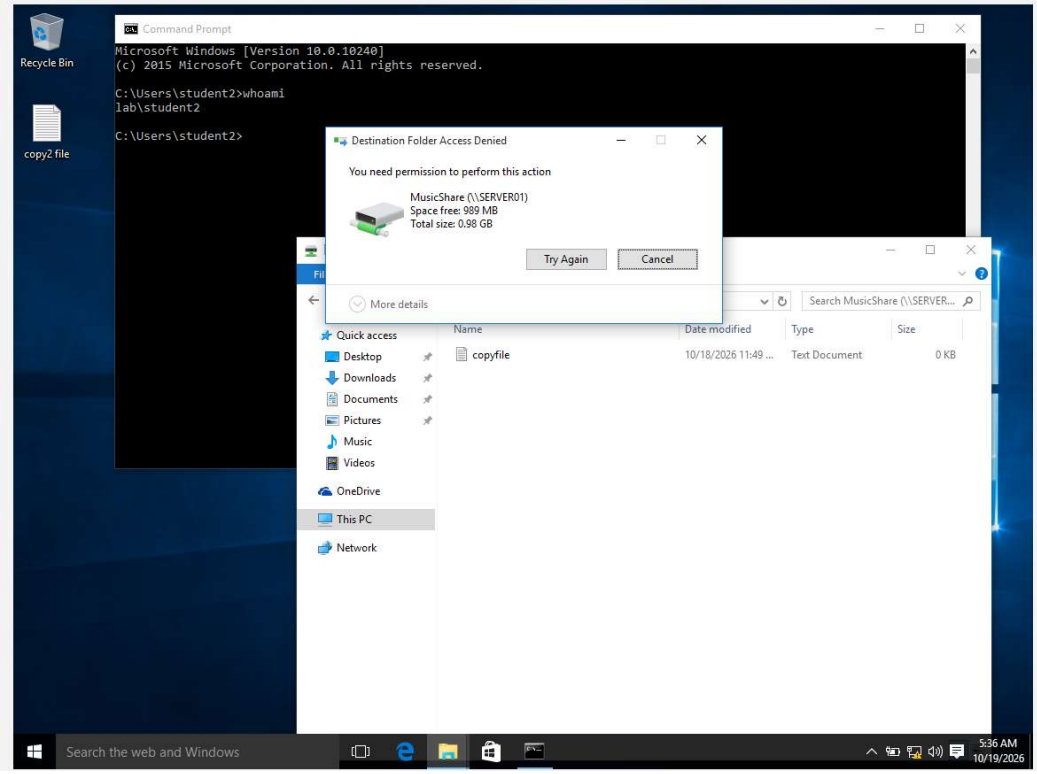


Figure 8: Logged in as lab\student2 (whoami); copy into the share denied

Results summary

User	Action	Result	Reason
Student 1	Copy file into share	Allowed	Share: Change (MusicLovers) and NTFS: Modify
Student 2	Delete file from share	Denied	Share allows it, but NTFS is Read-only; the stricter permission applies
Student 2	Copy file into share	Denied	NTFS has no Write permission for Student 2